

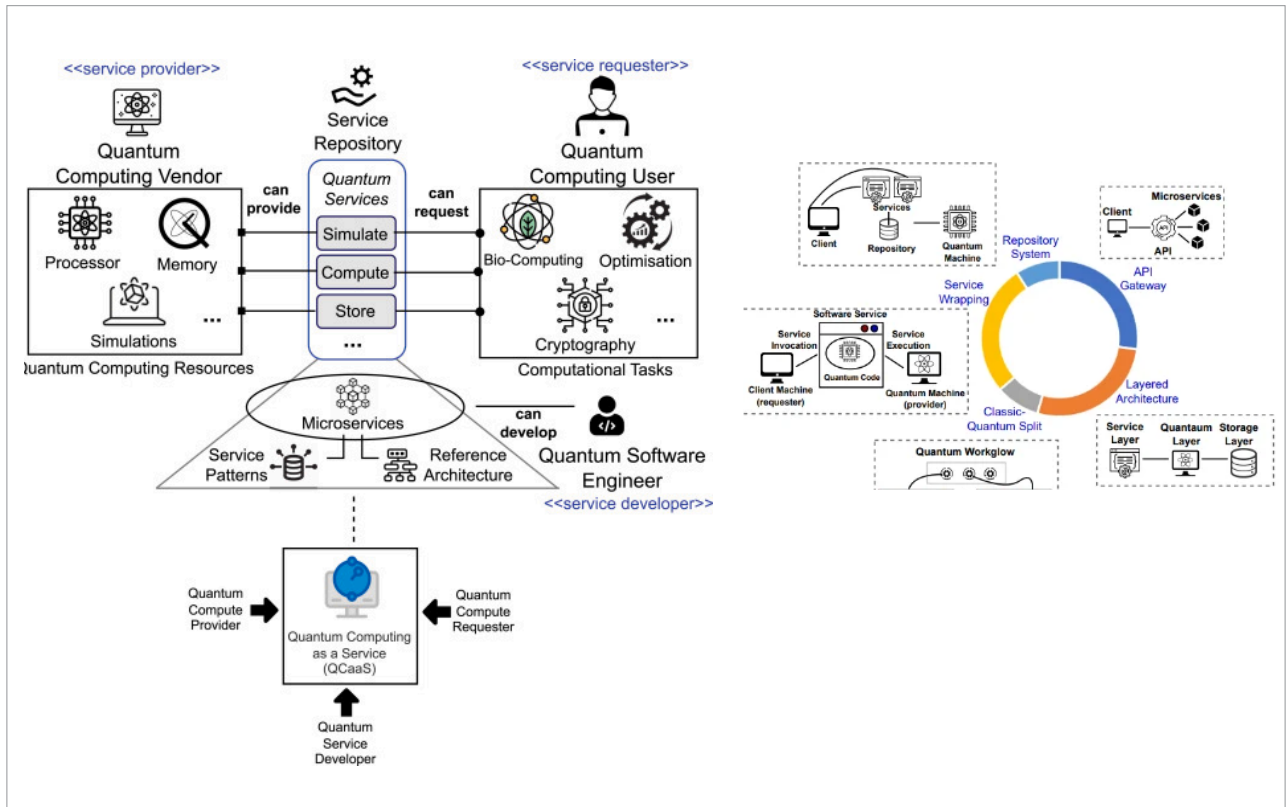


Figure 4: Quantum service computing

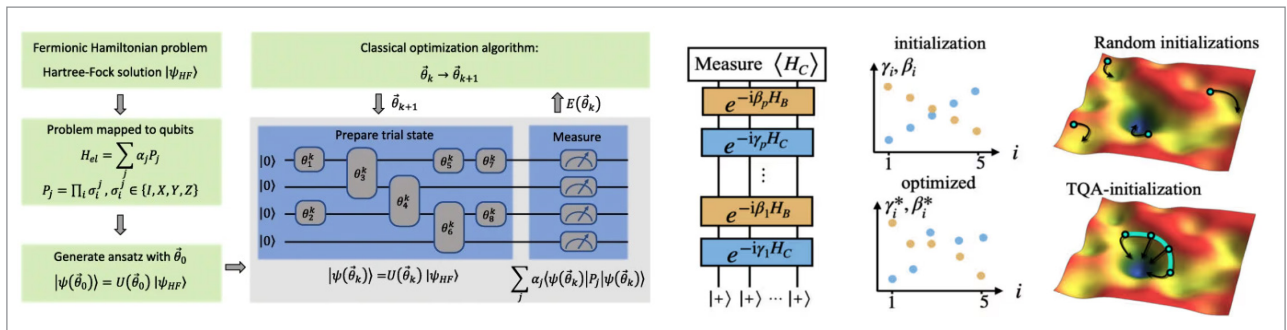


Figure 5: Variational Quantum Eigensolvers algorithm

- **Cloud orchestration layer:** This includes job scheduling and authentication.
- **Classical processing layer:** This includes classical processing.
- **Quantum hardware layer:** This includes the actual quantum processors.

This layered architecture enables developers to integrate quantum computing into existing workflows without requiring any specialised

hardware infrastructure.

Hybrid quantum-classical infrastructure: Orchestrating HPC and quantum workloads

Despite its tremendous potential, quantum computing is not intended to replace classical computers altogether. Instead, the most promising areas of application for quantum technology are those that will use a hybrid model of computing. In a hybrid system,

classical computers can be used for tasks such as data preparation, optimisation, and control of algorithms. Quantum processors can be used for specific areas of the calculation where it is believed that a quantum computer may be useful. The results can then be passed back to the classical computers for further analysis. This is the case for the Variational Quantum Eigensolvers algorithm and the Quantum Approximate Optimization Algorithm